Missing and Unbalanced Examples

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Recall and choose your starting point

> Record your starting explanation for Missing and Unbalanced Examples before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Good Data, Bad Data

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Similarity

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Boundaries and 'Not Sure'

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Missing and Unbalanced Examples trace and record the first step you can explain without help.

Use W2: Separate files from distinct cases for Sampling and **coverage**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Missing and Unbalanced Examples explanation against the supplied facts before discussing it.

Use the worked case for Sampling and **coverage**. Proposed explanation: “More records always remove bias”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Audit a new sound-recognition claim

A fixed target has 20 invented clips: 5 quiet bells, 5 noisy bells, 5 quiet claps and 5 noisy claps. A tester selects 4 distinct quiet bells from a quiet-bell-only list; 3 are correctly labelled and 1 is wrong.

Write a supported result, name missing conditions and explain why the result is not established for all 20 clips. Propose an 8-clip plan that includes every condition, with no clip selected twice.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Missing and Unbalanced Examples target to a trace and a justified conclusion.

A **training** set has 18 daylight photos and two dark photos of one object. Copying the dark photos eight times balances the counts. Does it balance the variety? Design a better collection step.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Missing and Unbalanced Examples case and show the deciding evidence.

Training includes nine daylight leaf photos and one low-light photo.

> What intended-use gap is present?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A photo sorter trains on ten bright photos and no dim photos. What group of examples is missing? Design a test that reveals the gap.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Missing and Unbalanced Examples adds to Helpful Classroom Sorter and keep the evidence.

List normal, low-light, partly hidden, worn, and unusual items; mark which conditions are missing from the sorter data.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A **training** set has 18 daylight photos and two dark photos of one object. Copying the dark photos eight times balances the counts. Does it balance the variety? Design a better collection step.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Name target, frame and sample

> Name target, frame and sample. Apply the supplied Missing and Unbalanced Examples facts and keep your reasoning.

Use the 24-card investigation. State the target collection, the original six-card **sample** and the three absent conditions.

If the selection list contains only bright-circle IDs, explain why random selection from that list cannot reach every target condition.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Separate files from distinct cases

> Separate files from distinct cases. Apply the supplied Missing and Unbalanced Examples facts and keep your reasoning.

Each of the six original **sample**d cards is photographed twice. State the number of image files and distinct cards. Has this covered a new shape or lighting condition? Explain why the counts need different labels.

> **My reasoning and evidence**

W3 Read a coverage table in combinations

> Read a coverage table in combinations. Apply the supplied Missing and Unbalanced Examples facts and keep your reasoning.

A different test includes 3 bright circles and 3 dim squares. A designer says it covers both shapes and both lighting conditions, so no combinations are missing.

Identify the missing combinations and propose six additional distinct cards to investigate them.

> **My reasoning and evidence**

WORKBOOK · 15 OF 21

W4 Audit a new sound-recognition claim

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Audit a new sound-recognition claim. Apply the supplied Missing and Unbalanced Examples facts and keep your reasoning.

A fixed target has 20 invented clips: 5 quiet bells, 5 noisy bells, 5 quiet claps and 5 noisy claps. A tester selects 4 distinct quiet bells from a quiet-bell-only list; 3 are correctly labelled and 1 is wrong.

Write a supported result, name missing conditions and explain why the result is not established for all 20 clips. Propose an 8-clip plan that includes every condition, with no clip selected twice.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Report a selected case without an outcome

> Report a selected case without an outcome. Apply the supplied Missing and Unbalanced Examples facts and keep your reasoning.

An 8-clip plan selects 2 from each sound condition. One selected noisy-clap file cannot be opened. The task measures recognition correctness only. State selected and assessed counts and the assessed count for each condition.

Explain why a report of all 8 checked is wrong, and how to handle the missing file transparently.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Missing and Unbalanced Examples

> A fictional badge reader will be used in both bright and dim rooms. Explain the labelled step

> before checking the changed condition.

A fictional badge reader will be used in both bright and dim rooms.

Intended target has 8 bright plain,8 bright patterned,8 dim plain,8 dim patterned cards.

A development **sample** contains 6 bright plain cards and 2 bright patterned cards; no dim cards.

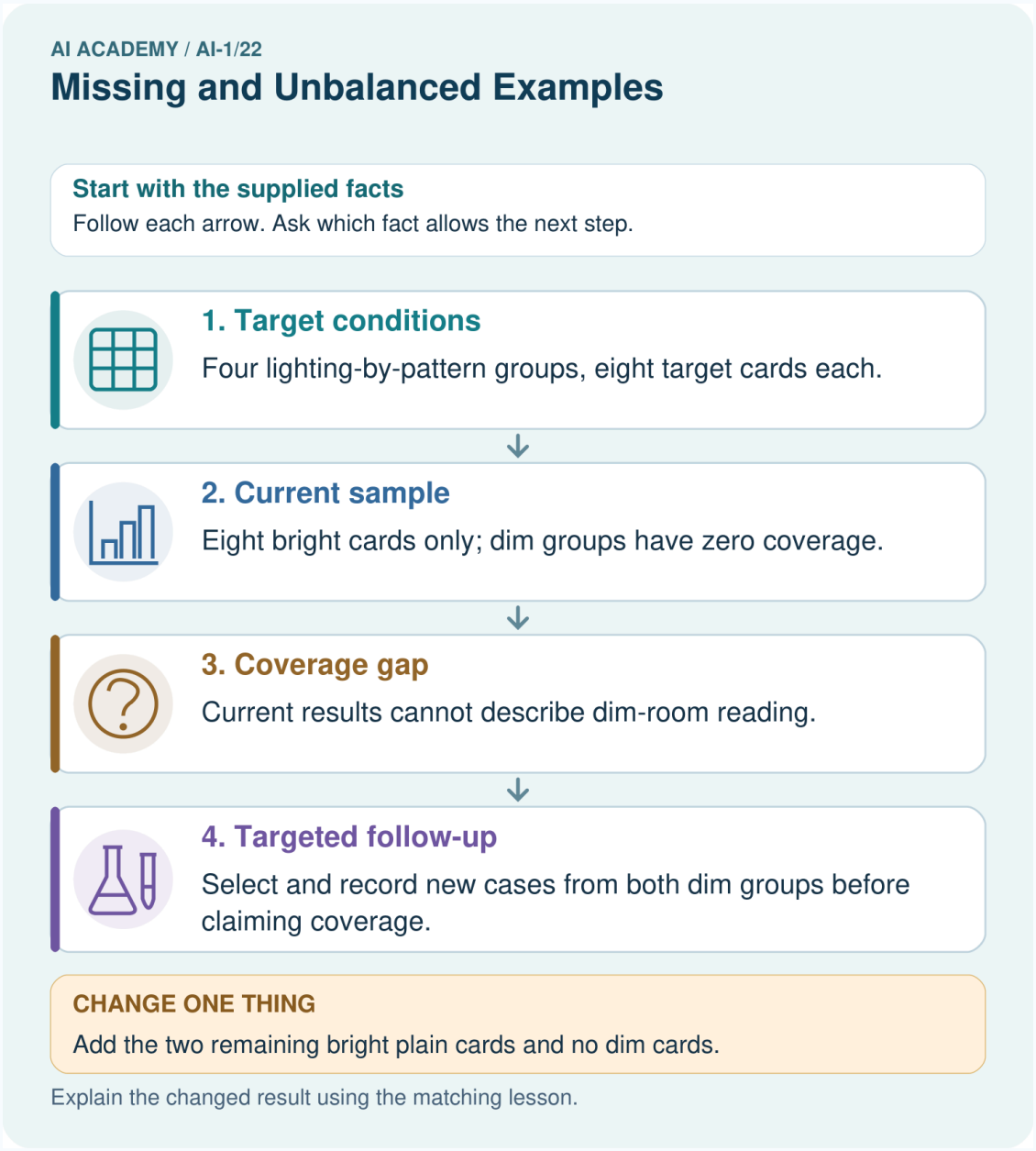
AI ACADEMY / AI-1/22

Target groups and development coverage

	Target cards	Sample cards
Bright plain	8	6
Bright patterned	8	2
Dim plain	8	0
Dim patterned	8	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Would testing only dim plain cards close every current gap? Point to the target groups.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Create a sorter **coverage** table crossing important classroom conditions, then state which groups each test actually represents.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Create a sorter **coverage** table crossing important classroom conditions, then state which groups each test actually represents.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional badge reader will be used in both bright and dim rooms.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Select and record new cases from both dim groups before claiming **coverage**.

1. Target conditions

2. Current **sample**

3. Targeted follow-up

4. **Coverage gap**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Targeted follow-up Select and record new cases from both dim groups before claiming **coverage**.

Adding more bright cards can increase **sample** size without addressing the missing condition. A **coverage** check asks which relevant situations are absent, not merely how many rows exist.

Deliberately selecting dim cases helps investigate those conditions, but the resulting test should not be advertised as a random survey unless its selection really was random.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add the two remaining bright plain cards and no dim cards. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The **sample** becomes larger but still has no dim **coverage**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The **sample** becomes larger

but still has no dim **coverage**. The missing lighting condition remains untested.

Repair a missing building block

Review Good Data, Bad Data (ai-1/21). Goal: Check records against a stated format, unit and meaning before using them. Short practice: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?.

Return to this week and explain what you changed.

Review Similarity (ai-1/19). Goal: Calculate similarity using explicitly selected features and a stated scoring rule. Short practice: Point to the matches supporting C's new score and explain why there is no tie..

Return to this week and explain what you changed.

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a **prediction**. Short practice: Which original block needs a better width before accepting or rejecting it?

What should the reviewer do?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

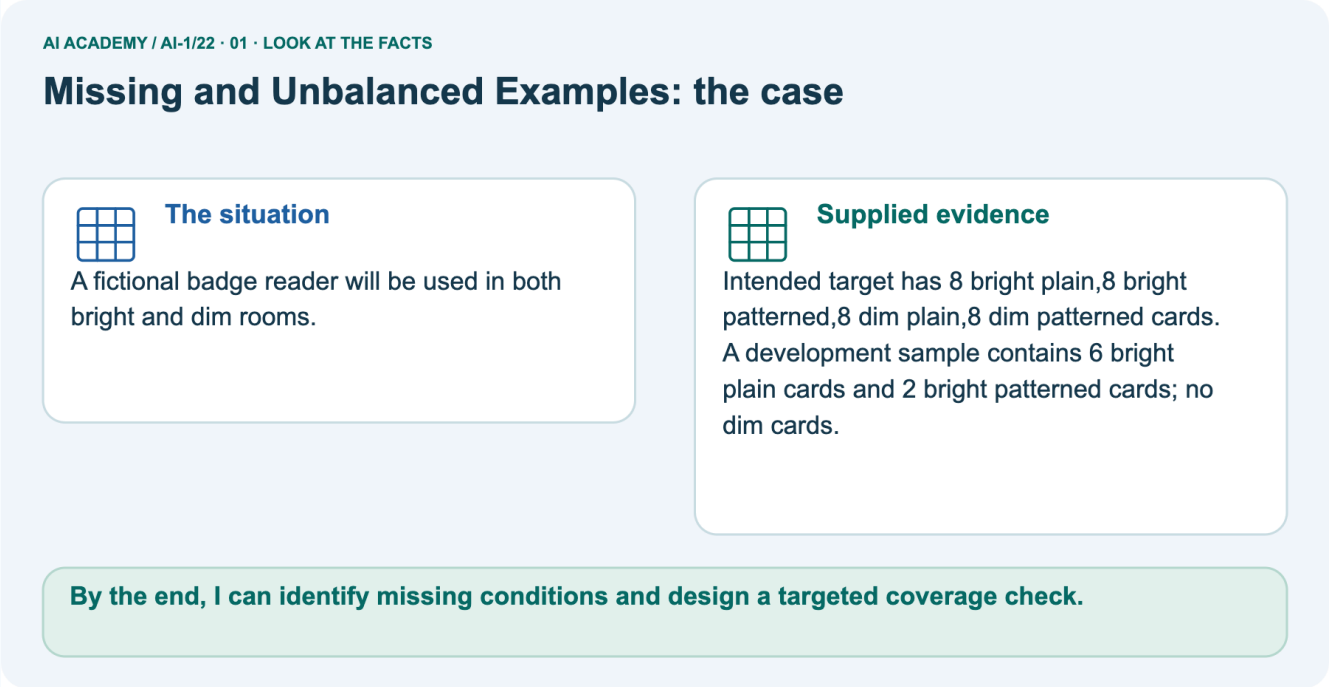
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Missing and Unbalanced Examples picture from its labels, then record a relevant limit.

01. Missing and Unbalanced Examples: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

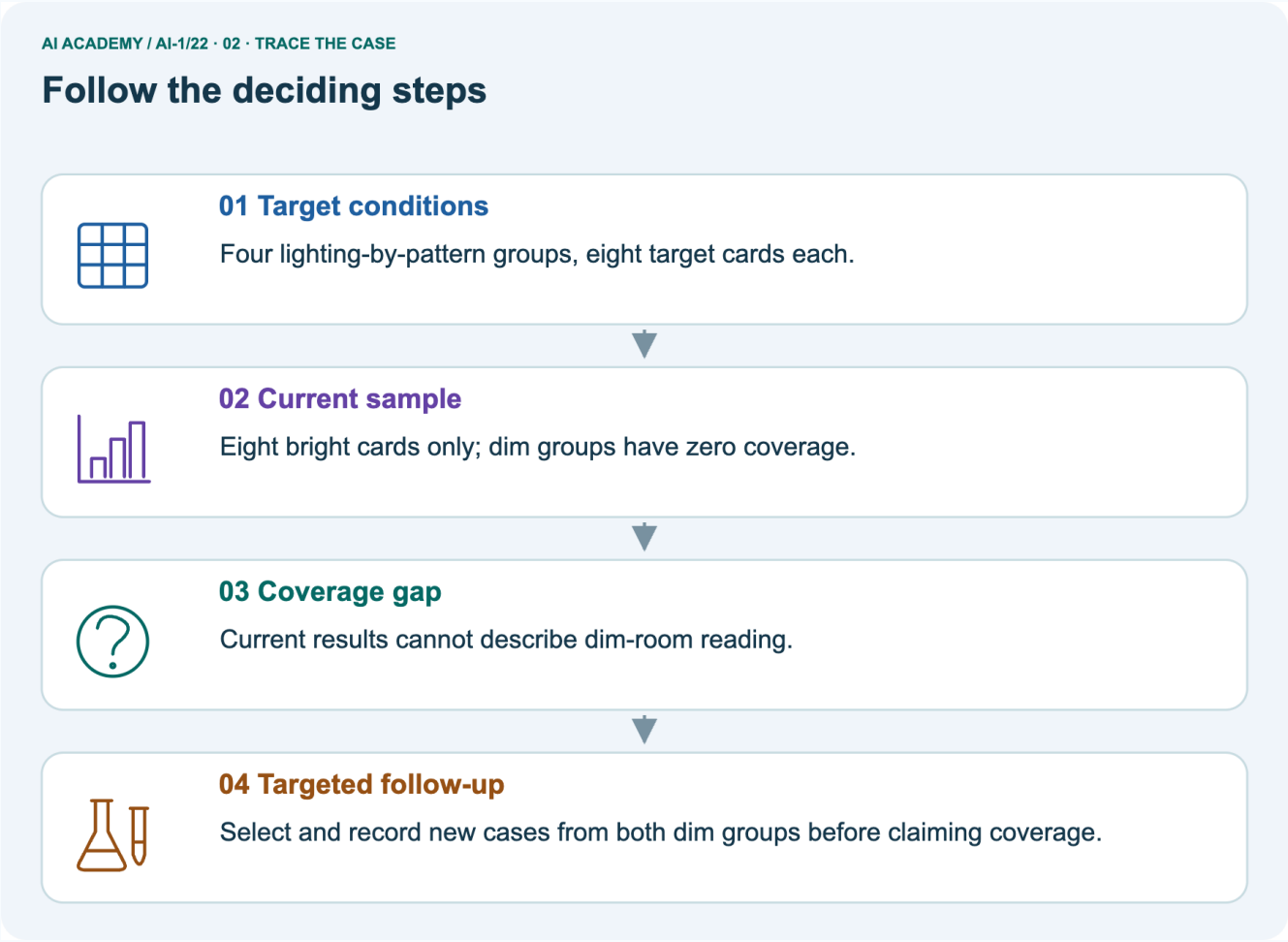
A fictional badge reader will be used in both bright and dim rooms. Supplied facts: Intended target has 8 bright plain,8 bright patterned,8 dim plain,8 dim patterned cards.

A development **sample** contains 6 bright plain cards and 2 bright patterned cards; no dim cards.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Target conditions: Four lighting-by-pattern groups, eight target cards each. → Current **sample**: Eight bright cards only; dim groups have zero **coverage**. → **Coverage gap**: Current results cannot describe dim-room reading. → Targeted follow-up: Select and record new cases from both dim groups before claiming **coverage**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Target groups and development coverage



Follow the labelled picture with the matching explanation. Open the vector diagram

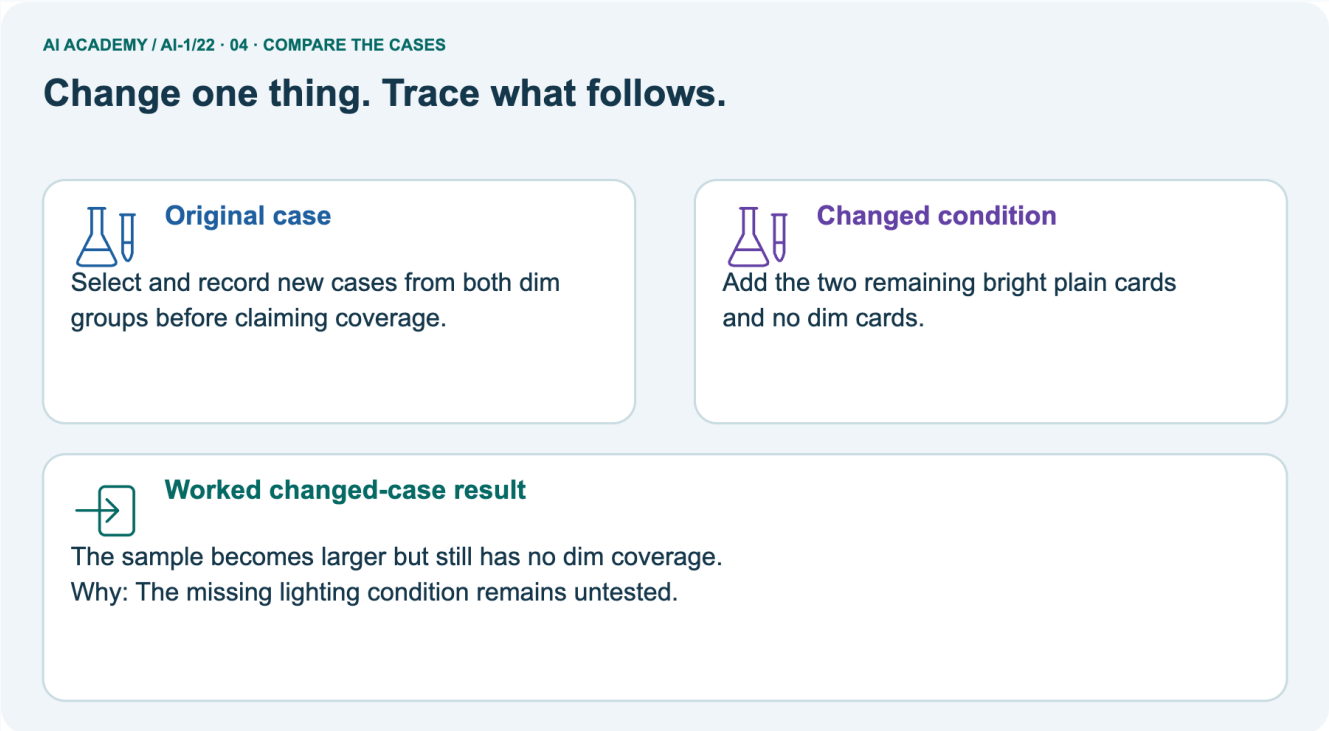
Target groups and development **coverage**. Bright plain: Target cards = 8, **Sample** cards = 6. Bright patterned: Target cards = 8, **Sample** cards = 2. Dim plain: Target cards = 8, **Sample** cards = 0.

Dim patterned: Target cards = 8, **Sample** cards = 0.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

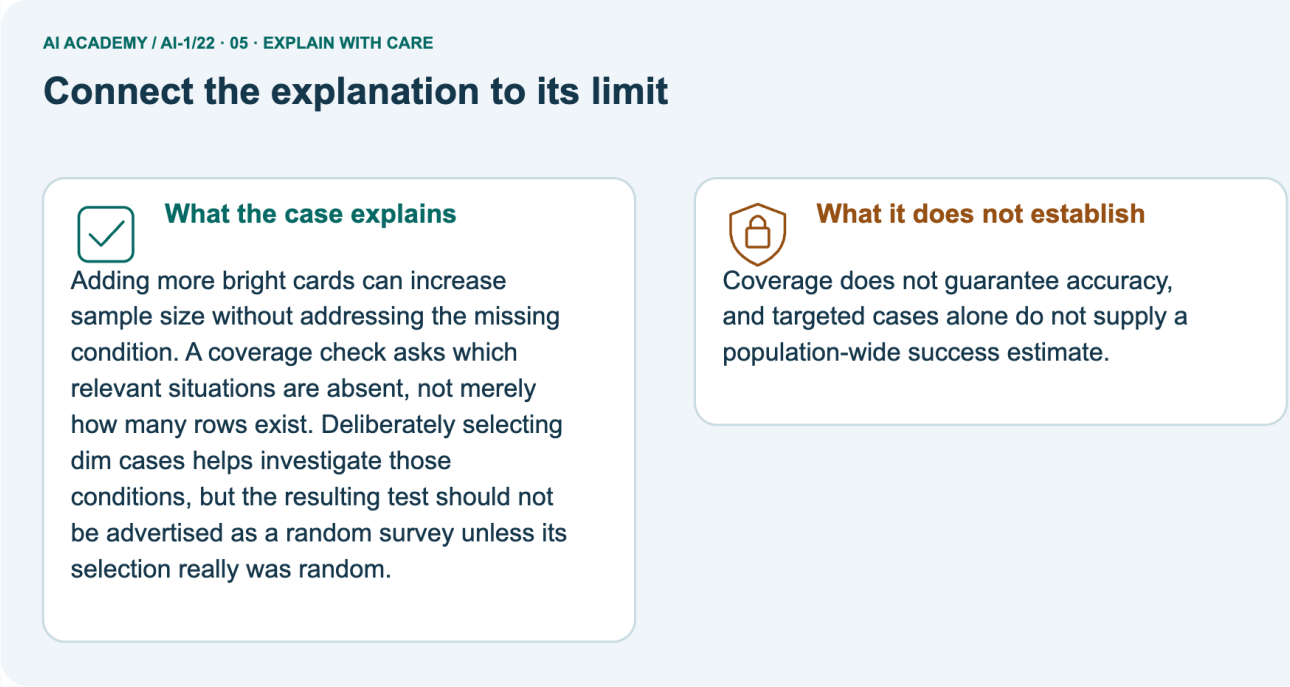
Original case: Select and record new cases from both dim groups before claiming **coverage**.
Changed condition: Add the two remaining bright plain cards and no dim cards.

Expected result: The **sample** becomes larger but still has no dim **coverage**. Reason: The missing lighting condition remains untested.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Adding more bright cards can increase **sample** size without addressing the missing condition. A **coverage** check asks which relevant situations are absent, not merely how many rows exist.

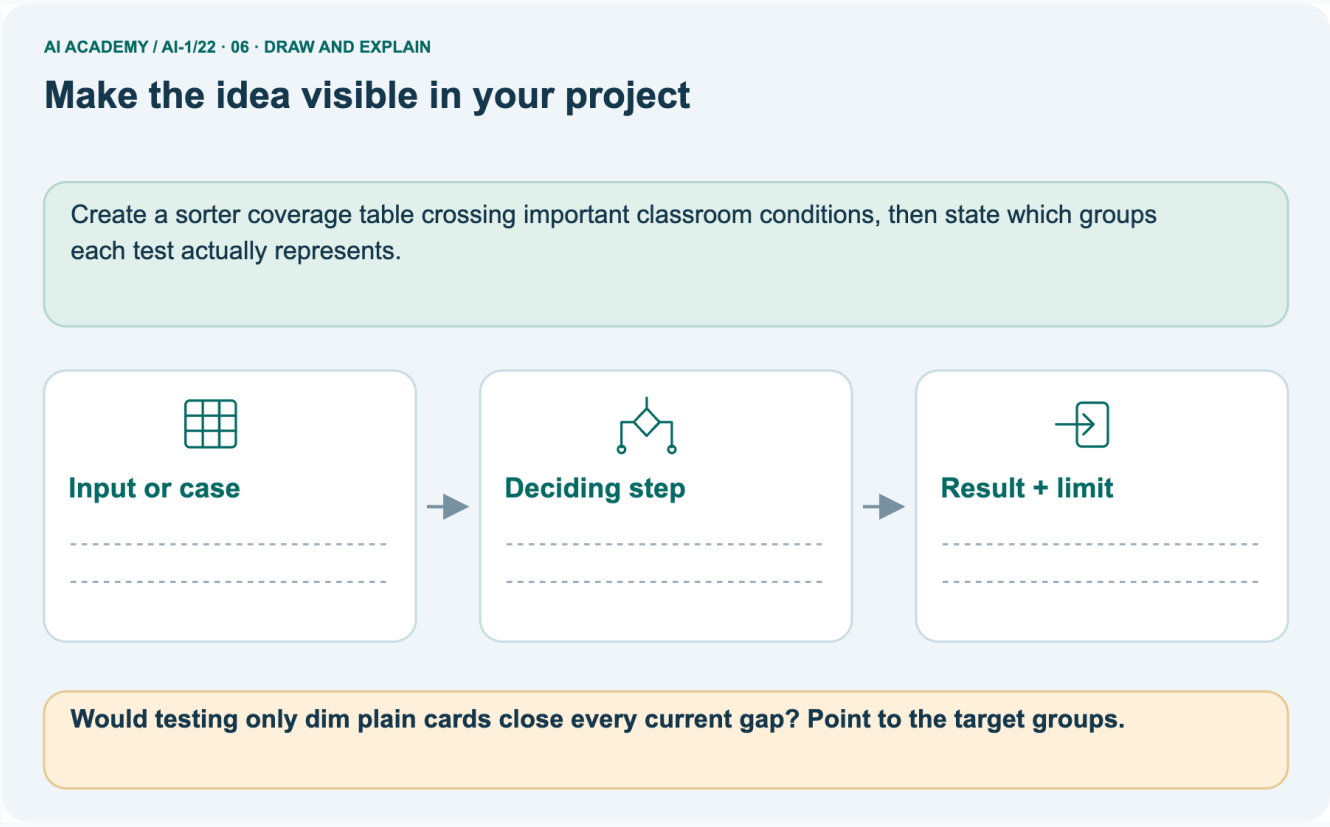
Deliberately selecting dim cases helps investigate those conditions, but the resulting test should not be advertised as a random survey unless its selection really was random.

Limit: **Coverage** does not guarantee accuracy, and targeted cases alone do not supply a population-wide success estimate.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create a sorter **coverage** table crossing important classroom conditions, then state which groups each test actually represents. Draw your **input**, deciding step, result and limit.

Supported question: Would testing only dim plain cards close every current gap? Point to the target groups.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.